

Introduction to Probability Theory

Probability, Random Variables, and Distributions- IE 1070

Conditional Probability, Independence, and Law of Total Probability

Mean value:

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

The Sample Variance:

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n - 1}$$

A permutation is an arrangement of all or part of a set of objects. The number of permutations of n distinct objects taken r at a time is

$${}_nP_r = \frac{n!}{(n-r)!}$$

Conditional Probability and Independence

When event B is given, the probability of A is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

An informal method to check the independence between two events is that if $P(A|B) = P(A)$, then they are independent. Generally, events $A_1, A_2, \dots, A_n$ such that

- $P(A_1 \cap \cdots \cap A_n) = P(A_1) \cdots P(A_n)$
 - Any non-empty subset of $\{A_1, A_2, \dots, A_n\}$ is independent
- are independent. For two events, A and B is independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$.
For those are dependent, we have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Partition and Total Probability

If a sample space is divided into some number of mutually exclusive events, and that their union makes up the whole sample space, it's called "collectively exhaustive" or a partition. A partition is countable sets $\{A_1, A_2, \dots, A_i\}$ in sample space S such that

- if $i \neq j$, then $A_i \cap A_j = \emptyset$
- $A_1 \cup A_2 \cup \dots \cup A_i = S$

For any event $B \in S$, we have the total probability formula:

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$$

Combination:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

The number of partitions can be calculated by following theorem:

Theorem: The number of ways of partitioning a set of n objects into r parts with $n_1, n_2, \dots, n_r$ elements is:

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}$$

Bayes' Theory

According to conditional probability, we can derive the equation with the assistance of $P(A \cap B)$:

$$P(B|A) \cdot P(A) = P(A|B) \cdot P(B)$$

Therefore, if $P(B) \neq 0$,

$$P(A|B) = P(B|A) \cdot \frac{P(A)}{P(B)}$$

The key point in this process is **exchangeability**: $A \cap B = B \cap A$.

Another critical point of Bayes's theory is that altering posterior probabilities based on new evidence and prior probabilities. We can express it by:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Discrete Random Variables

A **discrete random variable** is a real-value function defined in a discrete sample space Ω (which means Ω is finite or countable). Specifically, we assign a real number $X(\omega)$ to each element $\omega \in \Omega$.

Probability Mass Function (pmf) and Cumulative Distribution Function (cdf)

For discrete random variables X for all possible values x , its probability mass function is a set of ordered pairs $(x, f(x))$ such that

- $f(x) \geq 0$ for $\forall x$
- $\sum_x f(x) = 1$
- $P(X = x) = f(x)$

Cumulative distribution function of a discrete random variable $F(x)$ denotes the probability that $X \leq x$. The cdf can be given by pmf as

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

For discrete random variables, we have $P(a < X \leq b) = F(b) - F(a)$,
 $P(a \leq X \leq b) = F(b) - F(a) + f(a)$. From the definition we have:

$$f(x_i) = F(x_i) - F(x_{i-1})$$

Continuous Random Variables

Similarly, a **continuous random variable** is a real-value function defined in a continuous sample space Ω . The **Probability Density Function** is a real value function such that

- $f(x) \geq 0$, for all $x \in R$
- $\int_{-\infty}^{\infty} f(x)dx = 1$
- $P(a < X < b) = \int_a^b f(x)dx$

For continuous random variables, the cdf is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

From the definition we can see that:

$$P(a < X < b) = F(b) - F(a)$$

$$f(x) = \frac{dF(x)}{dx}$$

Joint Probability Distribution Function

For 2 discrete random variables X and Y , the *pmf* is defined as a function such that:

- $f(x, y) \geq 0$
- $\sum_x \sum_y f(x, y) = 1$

- $f(x, y) = P(X = x, Y = y)$

For any region A in the xy plane, $P((X, Y) \in A) = \sum \sum_A f(x, y)$.

The *pmf* of X and Y are given by summing another variables.

$$g(x) = \sum_y f(x, y), \quad h(y) = \sum_x f(x, y)$$

They are called **Marginal Probability Mass Function**.

For condition probability that $P(X = x|Y = y)$ is given by:

$$p(x|y) = \frac{f(x, y)}{h(y)}$$

The $p(x|y)$ is the conditional distribution of discrete random variable X given $Y = y$.

Similarly, for continuous random variables, the joint probability distribution function satisfies:

- $f(x, y) \geq 0$
- $\int_y \int_x f(x, y) \, dx dy = 1$
- $\iint_A f(x, y) \, dx dy = P([x, y] \in A)$

The marginal density equation is given by:

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Mathematical Expectation

Introduction to Expectation

Expectation is weighted average of all possible values of random variable. For discontinuous random variable, the expectation is defined as:

$$\mu(x) = E(x) = \sum_x x f(x)$$

Similarly, for continuous random variable, the expectation is

$$\mu(x) = E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

Also for the function of random variable $g(x)$, we have:

$$\mu(x) = E(x) = \sum_x g(x) f(x)$$

$$\mu(g(x)) = E(g(x)) = \int_{-\infty}^{\infty} g(x) f(x) \, dx$$

Moment of Random Variables

When $g(x) = x^r$, the expectation of $E(x^r)$ is r-order moment, $E((x - \mu)^r)$ is r-order center moment.

$$\mu_k = E[(x - \mu)^r] = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^r$$

The expectation is the first-order moment, and variance is the second-order center moment. The third-order center moment is skewness, which represents the asymmetry of the distribution. kurtosis is the forth-order center moment, which indicates the peak of the PDF at the mean value. For those two center moment, we have the following inequality

$$\frac{\mu_4}{\sigma^4} \geq \left(\frac{\mu_3}{\sigma^3} \right)^2 + 1$$

Properties of Expectation

If X and Y follows joint mass distribution $f(x, y)$, then

$$E(g(x, y)) = \sum_y \sum_x g(x, y) f(x, y)$$

If $f(x, y)$ is a joint density distribution, we have:

$$E(g(x, y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx dy$$

calculation of expectation is linear, which means

$$E(aX + bY) = aE(X) + bE(Y)$$

If two random variables are **independent**, then

$$E(XY) = E(X)E(Y)$$

Variance and Covariance

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum (x - \mu)^2 f(x)$$

if X is discrete, and

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx$$

if X is continuous. For two discrete random variables X and Y , the covariance is defined as

$$\sigma_{XY} = \text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are continuous, we have

$$\sigma_{XY} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$$

Generally, we have

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y$$

Covariance measures how a change in one variable affects a change in another variable, i.e., the degree of linear dependence. Therefore, we can define the correlation coefficient by standardize the covariance.

$$\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Properties of Variance

For any random variable, The variance can be calculated by

$$\text{Var}(X) = E(X^2) - E(X)^2$$

If new variable $Y = aX + b$, we have the variance

$$\sigma_Y^2 = a^2 \sigma_X^2$$

For any random variable $X = X_1 + X_2 + X_3 + \dots + X_n$, we have

$$\mu_X = \sum_{i=1}^n \mu_{X_i}$$

If they are independent, we know that

$$\sigma_X^2 = \sum_{i=1}^n \sigma_{X_i}^2$$

Discrete Distribution

Bernoulli distribution

If a random variable X satisfies $P(X = 1) = p$ and $P(X = 0) = 1 - p$, then X follows the Bernoulli distribution with parameter p , $X \sim \text{Bern}(p)$. We know that

$$\mu = p, \quad \sigma^2 = p(1 - p)$$

Binomial Distribution

The number of successes in n Bernoulli trial follows the **binomial distribution**

$$P(X = x) = b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

The mean and variance of the binomial distribution $b(x; n, p)$ are

$$\mu = np, \quad \sigma^2 = np(1 - p)$$

Multinomial Distribution

The binomial experiment becomes a multinomial experiment if we let each trial have more than two possible outcomes. Suppose that we have n trial with k possible outcomes with probability $p_1, p_2, p_3, \dots, p_k$. Then we have the joint probability distribution

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k p_i^{x_i}$$

where $\sum_{i=1}^k x_i = n$. Similarly, we have corollary like binomial distribution

$$\mu_{X_i} = np_i, \quad \sigma_{X_i}^2 = np_i(1 - p_i)$$

That's because for each possible outcome, the experiment can be described as a binomial experiment.

Hypergeometric Distribution

Let X be a random variable, the number of successes in n samples selected from N items. Then it follows the hypergeometric distribution. The probability distribution is given by

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad \max\{0, n - (N - k)\} \leq x \leq \min\{n, k\}$$

The mean value and variance is

$$\mu = \frac{nk}{N}, \quad \sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right)$$

When n is small compare to N , then the hypergeometric distribution approaches the binomial distribution.

For **Multivariate Hypergeometric Distribution**, we have

$$f(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n) = \frac{\binom{a_1}{x_1} \binom{a_2}{x_2} \dots \binom{a_k}{x_k}}{\binom{N}{n}}$$

where $\sum_{i=1}^k x_i = n$ and $\sum_{i=1}^k a_i = N$.

Negative binomial Distribution

Let X denote the number of trials that needed to get k success (with probability p) in a negative binomial experiment. The probability distribution of the random variable X , the number of the trial on which the k th success occurs, is

$$P(X = x) = b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x \geq k$$

Some ones will define the negative binomial distribution as the probability that the number X of successes when r th failures occur. Therefore

$$P(X = x) = \binom{k+r-1}{k} p^r q^k$$

with expectation $\frac{rp}{1-p}$ and variance $\frac{rp}{(1-p)^2}$.

Geometric Distribution

If repeated independent trials can result in a success with probability p and a failure with probability $q=1-p$, then the probability distribution of the random variable X , the number of the trial on which the **first** success occurs, is geometric distribution

$$P(X = x) = g(x; p) = pq^{x-1}, \quad x = 1, 2, 3, \dots$$

with expectation $\frac{1}{p}$ and variance $\frac{1-p}{p^2}$.

The cdf of geometric distribution is given by

$$F(x) = \sum_{y \leq x} P(Y = y) = p \sum_{y=0}^{x-1} (1-p)^y = 1 - (1-p)^x$$

Poisson Distribution

The Poisson distribution models the number of events occurring within a fixed interval of time or space, given that the events **happen with a constant rate** λ and are **independent** of each other.

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

The **mean** and **variance** of the Poisson distribution are **both** λ . For small values of λ , the distribution is highly skewed (leaning to the right). As λ increases, the distribution becomes more symmetric and approaches a normal distribution.

The Poisson distribution can be derived as a limit of the **binomial distribution**. Consider a scenario where we have:

- A large number of trials (e.g., very small intervals)
- A small probability of success in each trial (i.e., the rate λ is small)
- The number of trials n is large, but $p = \frac{\lambda}{n}$ is also small, so the expected number of successes in the interval remains constant at λ

As $n \rightarrow \infty$ and $p \rightarrow 0$, with $np = \lambda$, the binomial distribution approaches the Poisson distribution. This gives rise to the formula for the Poisson distribution

$$\text{Pois}(x) = \lim_{n \rightarrow \infty, np \rightarrow \lambda} \binom{n}{x} p^x q^{n-x} = \frac{\lambda^x e^{-\lambda}}{x!}$$

While $\lambda \rightarrow \infty$, the Poisson distribution approaches to the normal distribution.

Continuous Distribution

Flat Distribution

If the random variable X satisfies

$$P(X = x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

follows a flat distribution. Mean and variance is respectively

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$

Exponential Distribution

The exponential distribution is a continuous probability distribution often used to model the time between events in a **Poisson process**. A Poisson process is a type of stochastic process in which events occur randomly and independently at a constant average rate. The PDF is given by

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

λ is the rate parameter, which represents the **inverse of the mean time between events**. And the CDF is given by

$$F(x; \lambda) = 1 - e^{-\lambda x}, \quad x \geq 0$$

Mean and variance is respectively

$$\mu = \frac{1}{\lambda}, \quad \sigma^2 = \frac{1}{\lambda^2}$$

Note:

Normal Distribution

If PDF of an random variable X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

then it follows a normal distribution $X \sim N(\mu, \sigma^2)$, when $\mu = 0$ and $\sigma^2 = 1$ $N(0, 1)$ is the standard normal distribution. Z-score normalization $z = \frac{x-\mu}{\sigma}$ is used to perform standardization.

Normalization Constant Theorem

if there is a non-negative real function $g(x)$ such that

$$c = \int_{-\infty}^{\infty} g(x) \, dx$$

then $f(x) = \frac{g(x)}{c}$ is a PDF.

Properties

1. Symmetry

The normal distribution is **symmetric** about the mean μ . The curve is highest at μ and tapers off as moving away from the mean in either direction.

2. Bell-Shaped Curve

The graph of the normal distribution is a smooth, **bell-shaped curve**. It is the **Gaussian curve**, which is smooth, continuous, and unimodal (having a single peak at the mean).

3. 68-95-99.7 Rule

One of the most important properties of the normal distribution is the **68-95-99.7 rule**, which states:

- About **68%** of the data falls within one standard deviation of the mean, i.e., between $\mu - \sigma$ and $\mu + \sigma$.
- About **95%** of the data falls within two standard deviations of the mean, i.e., between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About **99.7%** of the data falls within three standard deviations of the mean, i.e., between $\mu - 3\sigma$ and $\mu + 3\sigma$.

4. Skewness and kurtosis

The **skewness** of a normal distribution is 0, meaning it is perfectly symmetric. The **kurtosis** of a normal distribution is **3** (often described as "mesokurtic"), meaning it has a moderate, bell-shaped peak.

5. Linearity

Any linear combination of independent normal random variables is also normally distributed.

If $X_1, X_2, \dots, X_n$ are independent normal random variables with means $\mu_1, \mu_2, \dots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, then for any constants $a_1, a_2, \dots, a_n$, the random variable:

$$Y = a_1X_1 + a_2X_2 + \dots + a_nX_n$$

is normally distributed with mean and variance given by:

$$\begin{aligned} E[Y] &= a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n \\ \text{Var}(Y) &= a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2 \end{aligned}$$

6. Cumulative Distribution Function (CDF)

The CDF of the standard normal distribution $N(0, 1)$ is denoted by

$$\Phi(x) = P(Z \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

The error function is defined as the probability that random variable $X \sim N(0, \frac{1}{2})$ within 0 to x .

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

It also can be calculated by the standard normal distribution and series expansion:

$$\Phi(x) - \Phi(-x) = \text{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)}$$

Approximate Binomial Distribution with Normal Distribution

The **normal distribution** can be used to approximate the **binomial distribution** to simplify the calculation.

Conditions for Approximation

- **Large number of trials:** n should be large enough. There is no strict cutoff, but a common rule of thumb is that the normal approximation works well when both np and $n(1-p)$ are greater than 5:

$$np \geq 5 \quad \text{and} \quad n(1-p) \geq 5$$

- **Moderate probability p :** The success probability p should not be too close to 0 or 1. If p is close to 0 or 1, the binomial distribution is highly skewed, and the normal approximation may not be accurate.

Approximation

For a random variable X following the binomial distribution with μ and σ^2 , we have approximation $X \sim N(\mu, \sigma^2)$. we can estimate the PDF and CDF by

$$P(X = k) = n \left(\frac{k - \mu}{\sigma} - 0.5 \leq z \leq \frac{k - \mu}{\sigma} + 0.5; \mu, \sigma \right)$$
$$P(X \leq x) = \sum_{r=0}^x b(r; n, p) = P(z \leq \frac{x + 0.5 - \mu}{\sigma})$$

where $x + 0.5$ is called continuity correction.

Continuity Correction

Since the binomial distribution is discrete and the normal distribution is continuous, a continuity correction is often applied to improve the approximation. This correction compensates for the difference between the discrete and continuous nature of the distributions. For a binomial random variable $X \sim \text{Binomial}(n, p)$, when using the normal approximation, the probability $P(X = k)$ is approximated by the probability $P(k - 0.5 \leq Z \leq k + 0.5)$ where $Z \sim N(np, \sqrt{np(1-p)})$.

For example, to approximate $P(X \leq k)$ for a binomial random variable,

$$P(X \leq k) \approx P\left(Z \leq \frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

Gamma Function

When s is not non-positive integer, the Gamma function is defined by the following integral:

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt = \int_0^{\infty} e^{-x} x^s \frac{dx}{x}$$

In fact, the integral converges where s is non-positive integer.

Analytic Continuation

Take the geometric series as an example. We know

$$S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n r^i = \frac{1}{1-r}$$

requires $|r| < 1$, but the RHS $y = \frac{1}{1-r}$ is defined for all $r \neq 1$. We called the RHS is an analytic continuation of LHS, and has a pole at $r = 1$. The Gamma Function can be extended by the recurrence relation

$$\Gamma(s+1) = s\Gamma(s)$$

which can be proved by partial integral. According to this relation, we can be inspired that

$$\Gamma(n + 1) = n!$$

Another way to extend the Gamma Function is using the functional equation reflection formula.

$$\Gamma(s)\Gamma(1 - s) = \pi \csc(\pi s) = \frac{\pi}{\sin(\pi s)}$$

Obviously, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, and for other half-integers, there are specific formulas that involve π , such as:

$$\Gamma\left(\frac{1}{2} + n\right) = \frac{(2n)!}{4^n(n!)^2} \cdot \sqrt{\pi}$$

Here are some special values:

- $\Gamma(1) = 1$
- $\Gamma(2) = 1! = 1$
- $\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$
- $\Gamma(3) = 2! = 2$

And there are also another way to study $\Gamma\left(\frac{1}{2}\right)$

Beta Function

The Beta function $B(x, y)$ is defined for $x > 0$ and $y > 0$ by the following integral

$$B(x, y) = \int_0^1 t^{x-1}(1 - t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x + y)}$$

From this function, we got another distribution: **Beta Distribution**.

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1} = \frac{x^{\alpha-1}(1 - x)^{\beta-1}}{B(\alpha, \beta)} \quad \text{for } x \in [0, 1]$$

With mean $\mu = \frac{\alpha}{\alpha + \beta}$ and variance $\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$. For $x = m$ and $y = n$ are positive integers, we have:

$$B(m, n) = \frac{(m - 1)!(n - 1)!}{(m + n - 1)!}$$

The shape parameters α and β determine the skewness and modality of the distribution:

- When $\alpha = \beta = 1$, the Beta distribution is uniform.
- When $\alpha > 1$ and $\beta > 1$, the distribution is unimodal and symmetric.
- When $\alpha < 1$ and $\beta < 1$, the distribution has U-shaped behavior, concentrating near 0 and 1.
- When $\alpha > \beta$, the distribution is skewed towards 1

- When $\alpha < \beta$, it is skewed towards 0.

Gamma Distribution and Normal Distribution

The $2m$ -order moment of standard normal distribution is given by

$$\mu_{2m} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^{2m} e^{-x^2/2} dx = \prod_{i=1}^m (2i-1) = \frac{2^m}{\sqrt{\pi}} \Gamma\left(m + \frac{1}{2}\right)$$

There is a connection between the **Gamma distribution** and the **normal distribution** in the context of large values of the shape parameter. Firstly, the PDF of Gamma Distribution is given by

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha) \beta^\alpha}, \quad x > 0$$

with mean $\mu = \alpha\beta$ and variance $\alpha\beta^2$. If α is large, the Gamma distribution can be approximated by a normal distribution

$$\text{Gamma}(\alpha, \beta) \sim N(\alpha\beta, \alpha\beta^2) \quad \text{as } \alpha \rightarrow \infty$$

Erlang Distribution

The Erlang distribution is a special case of the **gamma distribution**, where the shape parameter α is an integer. The PDF of the Erlang distribution is given by:

$$f(x) = \frac{x^{n-1} e^{-x/\beta}}{\beta^n (n-1)!}, \quad x > 0$$

It can be shown that if the times between successive events are independent, each having an exponential distribution with parameter β , then the total elapsed waiting time X until all n events occur has the Erlang distribution, with mean $\mu = \beta n$ and variance $\sigma^2 = \beta^2 n$.

Weibull Distribution

The PDF of the **Weibull Distribution** is given by

$$f(x; k, \lambda) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

with mean

$$\mu = \lambda \Gamma\left(1 + \frac{1}{k}\right)$$

and variance

$$\sigma^2 = \lambda^2 \left[\Gamma \left(1 + \frac{2}{k} \right) - \left(\Gamma \left(1 + \frac{1}{k} \right) \right)^2 \right]$$

when $k = 1$ it's **Exponential Distribution**, $k = 2$ it's **Rayleigh Distribution**. The **hazard function** $h(x)$, which gives the instantaneous failure rate at time x , is the ratio of the PDF to the survival function (1 minus the CDF). For the Weibull distribution:

$$h(x; k, \lambda) = \frac{f(x; k, \lambda)}{1 - F(x; k, \lambda)} = \frac{k}{\lambda} \left(\frac{x}{\lambda} \right)^{k-1}$$

The hazard function characterizes the likelihood of failure occurring at a given time. Depending on the value of k , the hazard function can either increase or decrease with time which reflects whether the failure rate is increasing or decreasing.

Cauchy Distribution

The Cauchy distribution is a continuous probability distribution that is notable for its heavy tails and lack of a well-defined mean and variance. It is often used in scenarios where data exhibit **extreme outliers** or **heavy-tailed behavior**. PDF is given by

$$f(x; x_0, \gamma) = \frac{1}{\pi \gamma \left[1 + \left(\frac{x - x_0}{\gamma} \right)^2 \right]}$$

for standard Cauchy distribution is

$$f(x) = \frac{1}{\pi} \frac{1}{1 + x^2}$$

and CDF:

$$F(x; x_0, \gamma) = \frac{1}{2} + \frac{1}{\pi} \arctan \left(\frac{x - x_0}{\gamma} \right)$$

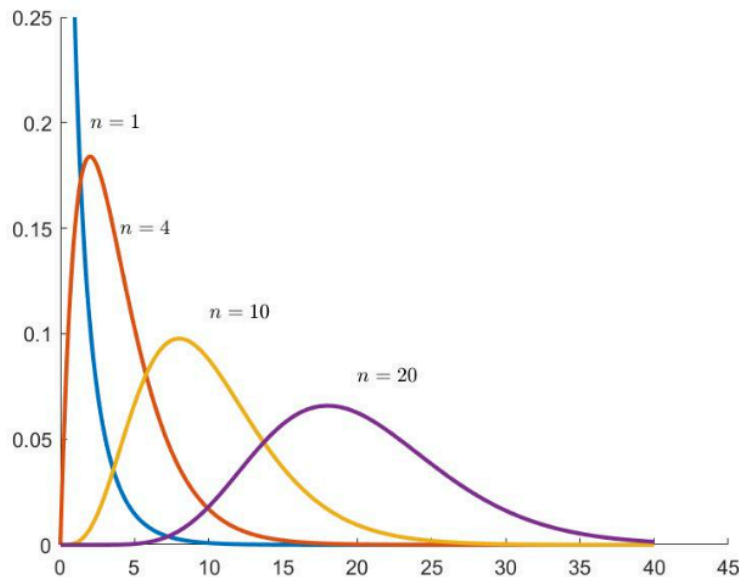
The Cauchy Distribution has no mean and variance.

Chi-Square Distribution

The PDF is given by

$$f(x; \nu) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\frac{\nu}{2}-1} e^{-x/2}, \quad x > 0, \nu > 0$$

with mean $\mu = \nu$, and variance $\sigma^2 = 2\nu$. Where ν is the degree of freedom, which determine the shape of the Chi-Square Distribution.



For k random variables X_i following the standard normal distribution $X_i \sim N(0, 1)$, then

$$Z = \sum_{i=1}^k X_i^2$$

follows the Chi-Square Distribution.

Limit Theorem

Inequality and Law of Large Numbers

Inequality

Markov's Inequality

Markov's inequality provides an upper bound on the probability that a non-negative random variable is greater than or equal to a certain value. It is one of the simplest and most widely used inequalities in probability theory.

Theorem: Let X be a non-negative random variable, and $a > 0$. Then:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's Inequality

Chebyshev's inequality gives a bound on the probability that a random variable deviates significantly from its mean. It is a generalization of the Law of Large Numbers.

Theorem: Let X be a random variable with expected value μ and variance σ^2 . For any $k > 0$, the following inequality holds:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Boole's Inequality and Bonferroni's Inequality

Boole's inequality, also known as the **Union Bound**, gives a bound on the probability of the union of events. It is a fundamental result in probability theory.

Theorem: Let $A_1, A_2, \dots, A_n$ be events in a probability space. Then:

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

The Bonferroni's inequality says that, for positive integer l, m , we have

$$\sum_{k=1}^{2l} (-1)^{k-1} S_k \leq P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{k=1}^{2m-1} (-1)^{k-1} S_k$$

where S_k is the sum of probability of all possible intersection of k events A_i

$$S_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k})$$

Convergence

Convergence in Distribution

It's the most **weak convergence** and contained in every other types of convergence. A

sequence of random variables $X_1, X_2, \dots$ converges in distribution (denoted as $X_n \xrightarrow{d} X$) to a random variable X if the cumulative distribution function (CDF) of X_n converges to the CDF of X at all points where F_X is continuous:

$$F_n(x) \rightarrow F_X(x) \quad \text{for all } x \in \mathbb{R}$$

This is also known as weak convergence or convergence in law.

Example: For uniform distribution $f(x) = \frac{1}{n}$, $x \in \{0, \frac{1}{n}, \dots, \frac{n-1}{n}\}$, the CDF is

$$F(x) = \frac{\lfloor nx \rfloor}{n}$$

because we know $nx - 1 \leq \lfloor nx \rfloor \leq nx$

$$x - \frac{1}{n} \leq F_n(x) \leq x$$

According to the Squeeze theorem,

$$\lim_{n \rightarrow \infty} F_n(x) = x$$

Convergence in Probability

A sequence of random variables $X_1, X_2, \dots$ converges in probability to a random variable X (denoted as $X_n \xrightarrow{P} X$) if, for every $\varepsilon > 0$, the probability that the absolute difference between X_n and X is greater than ε goes to zero as n approaches infinity:

$$P(|X_n - X| \geq \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Example: considering the constant random variable δ_c : $P(X = c) = 1$, and the normal distribution $N(c, \frac{1}{n^2})$. According to Chebyshev's inequality, we have

$$0 < P(|X_n - c| \geq k\sigma) \leq \frac{1}{k^2}$$

Because $\sigma = \frac{1}{n}$, let $k = n\varepsilon$.

$$P(|X_n - c| \geq \varepsilon) \leq \frac{1}{n^2\varepsilon^2}$$

when $n \rightarrow \infty$, $P(|X_n - c| \geq \varepsilon) \rightarrow 0$

Convergence Almost Surely and Surely

Considering a probability space $(\Omega, \mathcal{F}, P)$, a sequence of random variables $X_1, X_2, \dots$ converges almost surely (denoted as $X_n \xrightarrow{a.s.} X$) to a random variable X if:

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

That is, the sequence X_n converges to X with probability 1. If

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega), \quad \text{for all } \omega \in \Omega$$

then $\{X_n\}_{n=1}^{\infty}$ surely converges to X .

Law of Large Number

Let $X_1, X_2, \dots$ be a sequence of iid random variables. **The Weak Law of Large Numbers states** that for any $\epsilon > 0$, the probability that the sample mean $\bar{X}_n$ deviates from the true mean μ by more than ϵ tends to zero as the sample size n increases

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq \epsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

In other words, the sample mean converges to the true mean μ **in probability** as $n \rightarrow \infty$.

The Strong Law of Large Numbers is a stronger version of the weak law. It states that, under the same conditions, the sample mean X_n converges to the expected value μ almost

surely (with probability 1):

$$P\left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu\right) = 1$$

For weak law, large deviations from μ may occur with small probability, but they never disappear completely. For strong law, large deviations from μ happen with probability 0 after a certain point.

Stirling's approximation

While $n \rightarrow \infty$, the Stirling's approximation states that

$$n! \approx n^n e^{-n} \sqrt{2\pi n} \quad \text{or} \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n e^{-n} \sqrt{2\pi n}}$$

and we have the expansion

$$n! = n^n e^{-n} \sqrt{2\pi n} \left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} - \dots\right)$$

this series does not converge. It's an **asymptotic series**. Unlike a **convergent series**, an asymptotic series doesn't necessarily converge to the function for large values of the parameter. Instead, it gives an increasingly accurate approximation for a sufficiently large value of the parameter, even though the series may not converge in the traditional sense.

Convolution

The generating function for $\{a_n\}_{n=0}^{\infty}$ is defined as

$$G_a(s) = \sum_{n=0}^{\infty} a_n s^n$$

where s is any number that makes this series convergent. There are some types of generating functions. The **probability generating function** of a discrete random variable X takes the form of a power series that generates the probabilities of the outcomes of X .

$$G_X(s) = E[s^X] = \sum_{k=0}^{\infty} P(X = k) s^k$$

If $X_1, X_2, \dots, X_n$ are independent and non-negative random variables, and $Y = \sum_{i=1}^n X_i$, we have:

$$G_Y(s) = \prod_{i=1}^n G_{X_i}(s)$$

which means the PDF of the sum of independent random variables is the convolution of each PDF. If X is a continuous random variable, the the probability generating function is defined as

$$G_X(s) = \int_{-\infty}^{\infty} s^x f(x) dx$$

The Moment Generating Function (MGF) of a random variable X is defined as the expected value of the exponential function of the random variable X :

$$M_X(t) = E[e^{tX}] = \begin{cases} \sum_{i=0}^{\infty} e^{tx_i} f(x_i) & \text{discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & \text{continuous} \end{cases}$$

where t is a real (or complex) parameter, and X is a random variable. The function $M_X(t)$ is typically a function of t , and it is used to generate the moments of X .

Properties of MGF

Let μ_k be the moment of a random variable, we have

$$M_X(t) = 1 + \mu_1 t + \frac{\mu_2 t^2}{2!} + \dots = \sum_{n=0}^{\infty} \frac{\mu_n t^n}{n!}$$

and we can see that

$$\frac{d^n}{dt^n} M_X(0) = E(X^n) = \mu_n$$

The linearity of MGF is given by

$$M_{\alpha X + \beta}(t) = e^{\beta t} M_X(\alpha t)$$

and MGF is unique for discrete random variables. However, you can not say that to continuous RV. There is different distribution with the same moments.

Central Limit Theorem

Statement

Let $X_1, X_2, \dots, X_n$ be a sequence of iid random variables with mean μ and variance σ^2 , and let $S_n = X_1 + X_2 + \dots + X_n$ be their sum. Then, the Central Limit Theorem states that as $n \rightarrow \infty$, the distribution of the normalized sum approaches the standard normal distribution:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

and for continuous case, the distribution of **standardized sample mean** converges to the

standard normal distribution

$$\frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{d} N(0, 1)$$

Special case is for Poisson distribution with parameter λ , the sum converges to $N(\lambda, \lambda)$.